

PAIRS TRADING AND VECTOR AUTOREGRESSION
APPLIED TO ENERGY EQUITIES: A REPLICATION
AND EXTENSION IN R PROGRAMMING LANGUAGE

by

Silvanus Brian Mulu, B.A.

A Honors Thesis submitted to the Graduate Council of
Texas State University in partial fulfillment
of the requirements for the degree of
Bachelor of Arts
with a Major in Economics and Mathematics
May 2029

Supervisor: Dr. Anurag Deb

COPYRIGHT

by

Silvanus Brian Mulu

2029

DEDICATION

ACKNOWLEDGMENTS

I am grateful to my supervisor, Dr. Anurag Deb, who has taught me macro-econometric principles and guided me throughout my research period at Texas State University. My interest in this project stems from a curiosity about, and enthusiasm for, applying my Economics and Mathematics background to something I am genuinely passionate about, global financial markets. Working through the Vector Autoregression (VAR) in R has given me the opportunity to connect theoretical personal reading to an applied problem and this thesis is a living testimony to that.

TABLE OF CONTENTS

	Page
LIST OF TABLES	vii
LIST OF FIGURES	viii
LIST OF ABBREVIATIONS	ix
ABSTRACT	x
 CHAPTER	
1 INTRODUCTION	1
1.1 Purpose and scope of this paper	1
2 DATA	3
2.1 Sample	3
2.2 Data Source and Adjustment	4
2.3 Return Construction	5
2.4 Sample Period and Structural Context	5
3 EXPLORATORY DATA ANALYSIS	6
3.1 Descriptive statistics	6
3.1.1 Central tendency	6
3.1.2 Dispersion and volatility ordering	7
3.1.3 Skewness	7
3.1.4 Kurtosis and the comparison with the original study	8
3.1.5 The extreme observations in standard deviation units	9
3.2 Formal tests of normality	14
3.3 Standardised returns and Graphical Analysis	15
3.3.1 Density comparison	15
3.3.2 Quantile comparison	17
3.4 Correlation structure	18
4 VECTOR AUTOREGRESSION	20
4.1 The role of this section	20
4.2 Confirming the order of integration	20
4.3 VAR Theoretical Framework	23
4.3.1 Mathematical Formulation of the $VAR(p)$ Model	23
4.4 Stacked Matrix System Representation	25
4.5 Closed Form System Estimation via OLS	27

4.6	Residual Extraction and Innovation Covariance Estimation	28
4.7	Lag order selection	29
4.7.1	Graphical Analysis the Chosen Information Criteria	31
4.8	Estimation Results	32
4.8.1	Goodness of Fit and Predictive Capacity	32
4.8.2	Stability	33
4.8.3	Coefficient sign pattern and a microstructure caveat	35
4.8.4	Innovation covariance structure	36
APPENDIX SECTION		39
Bibliography		40

LIST OF TABLES

	Page
Table 3.1 Descriptive statistics of daily log returns, 2000 to 2026	6
Table 3.2 Annualised moments of daily log returns, 1999 to 2026	6
Table 3.3 Excess kurtosis, original sample against extended sample	8
Table 3.4 Worst observed daily return expressed in standard deviation units (z - scores).	9
Table 3.5 Gaussian tail probabilities and implied waiting times for worst ob- served returns.	12
Table 3.6 Normality tests on daily log returns	15
Table 3.7 Correlation matrix of daily log returns	18
Table 4.1 Augmented Dickey and Fuller tests on price levels. Lag length selected by AIC subject to the Schwert bound of 34.	21
Table 4.2 Augmented Dickey and Fuller tests on daily log returns.	21
Table 4.3 Kwiatkowski, Phillips, Schmidt and Shin tests on price levels. The null hypothesis is stationarity.	22
Table 4.4 Lag order selected by each Information Criterion, $p_{\max} = 20$	29
Table 4.5 VAR(2) Coefficient Estimates and Model Fit Statistics	32
Table 4.6 Contemporaneous correlation matrix of VAR(2) residuals.	37

LIST OF FIGURES

	Page
Figure 3.1 Log-density comparison of tail decay under Gaussian vs. heavy-tailed Student's t specifications.	14
Figure 3.2 Kernel density estimates of standardised daily log returns with a simulated standard normal $\mathcal{N}(0, 1)$ control	16
Figure 3.3 Normal Q-Q plots of standardized daily log returns versus theoretical Gaussian quantiles.	17
Figure 4.1 AIC, HQ and SC evaluated across lag orders $p \in \{1, \dots, 20\}$. The open circles highlight the global minimum of each Information Criteria.	31

ABSTRACT

This paper replicates the study by Salas Najera [1], which applies cointegration based pairs trading to two integrated oil majors, one exploration and two exchange traded funds. The original study is implemented in Python while this replication is implemented in R. This study adds approximately seven and a half years of new sample data, extending the original study's endpoint of December 2018 through July 2026. Therefore, my sample spans regime shifts such as the 2020 collapse in crude oil price, consolidation of the United States shale assets and the recent 2026 U.S. - Iran conflict [2].

1. INTRODUCTION

Pairs trading, also described as statistical arbitrage, was further developed at Morgan Stanley during the 1980s under the direction of Nunzio Tartaglia's quantitative research group [3]. The premise is that certain securities are bound together by economic fundamentals in a way that constrains how far their prices can drift apart. When the gap between two such securities diverges beyond its historical range, the strategy sells short the overperforming (expensive) leg and buys the relatively underperforming (cheap) leg. Profit is realised as the spread reverts to its mean, provided the underlying economic relationship between the two assets remains the same.

1.1. Purpose and scope of this paper

As a curious student pursuing Economics and Mathematics, I have developed a keen interest in the global financial markets and as an international student, the liquidity, volatility, market sentiments fast paced stock market has piqued my interest alot. I therefore embarked this journey as mere way to connect theoretical frameworks my thousands of pages to apply it in the financial market with a bit of creativity. Whilst learning VAR, I wanted to try do something and found the paper. This paper replicates the study by Salas Najera [1], a Certificate in Quantitative Finance project that applies the co-integration method to two integrated oil majors (XOM and CVX), two exchange traded funds (XLE and SPY) and one exploration and production company (COP). The original study is implemented in Python and covers 31 December 1999 to 31 December 2018. This replication is implemented independently in R and extends the sample forward to 24 July 2026.

The replication has two purposes. The first is pedagogical: reimplementing a study from its written description, not only in a different programming language but also unique point of view forces engagement with every methodological choice that a reader would otherwise skim past. The second is substantive. The original study ends in December 2018, so the

period from January 2019 onward is data that the original author could not have observed and could not have used, whether deliberately or inadvertently, in selecting the pair or tuning the strategy parameters. This is a genuine holdout sample rather than a simulated one and it happens to span an unusually severe set of market stresses. These include: the collapse of crude oil prices in March and April 2020, the subsequent recovery through 2022, geopolitical conflicts and Middle Eastern supply shocks and a wave of consolidation in United States shale that materially altered the composition of the very companies in this study.

2. DATA

2.1. Sample

The sample consists of three common stocks and two exchange traded funds observed at daily frequency from 31 December 1999 to 24 July 2026, giving 6,680 price observations and 6,679 return observations per series with no missing values.

The three stocks are ConocoPhillips (COP), Chevron Corporation (CVX) and Exxon Mobil Corporation (XOM). While Chevron and ExxonMobil are integrated majors operating across exploration, refining and distribution, ConocoPhillips is a pure-play exploration and production (E&P) company following its 2012 downstream spin-off [4]. This structural distinction in business models matters when evaluating how refining margins versus pure upstream commodity exposure impact the pair’s co-integration. A pure exploration and production firm is levered almost entirely to the crude price, whereas a refiner benefits from a wide crack spread and can profit when crude falls. Integrated firms carry both exposures, so restricting the sample to integrated majors removes an obvious source of structural divergence between the legs of a candidate pair. The three are also the largest United States energy companies by market capitalisation, which means they share a capitalisation classification, are held by broadly the same institutional investor base and also trade with sufficient liquidity that a spread strategy could realistically be executed without the bid ask spread consuming the signal.

Two exchange traded funds are included for context rather than as primary trading candidates. The Energy Select Sector SPDR Fund (XLE) tracks the energy component of the S&P 500 and serves as the sector benchmark against which strategy performance is later judged. The SPDR S&P 500 ETF Trust (SPY) tracks the broad index and represents the aggregate market factor.

2.2. Data Source and Adjustment

Prices were retrieved from Yahoo Finance through the `quantmod` package in R, which follows the standard CRSP-style adjustment methodology [5]. The adjusted close series is used throughout rather than the raw close.

This was done as the adjusted close removes the mechanical effects of stock splits and cash dividends. When a company executes a split, the raw price falls by the split ratio overnight even though no shareholder has gained or lost anything, and an unadjusted return series would record a large spurious loss on that date. Similarly, on an ex dividend date the raw price falls by approximately the dividend amount, because that cash has left the firm, yet the shareholder is unharmed since they receive the cash directly. The adjusted series folds both effects back into the historical path, so that the resulting return series measures the total return actually experienced by a holder of the position rather than price appreciation alone.

For this sample this distinction is very important. Integrated oil majors have historically paid dividend yields in the region of 3% - 5%, so over 26 years the cumulative difference between price return and total return is large. The adjusted price of ConocoPhillips on 31 December 1999 is 7.73 against a nominal traded price several times higher, and the entire difference is accumulated distributions and splits. More importantly for the analysis that follows, an unadjusted series would embed a downward jump in each leg on every ex dividend date. Because the two legs of a pair pay dividends on different calendars, those jumps would not offset within the spread and would appear as trading signals that correspond to no economic dislocation whatsoever. The original study does not state which series it uses. This replication uses adjusted close and notes the choice as a point of possible divergence.

2.3. Return Construction

Returns are computed as continuously compounded log returns,

$$r_t = \ln P_t - \ln P_{t-1} \tag{2.1}$$

where P_t denotes the adjusted close on day t . Log returns are preferred to simple returns for two reasons. They are additive across time, so that the return over any horizon is the sum of the daily returns within it. They are also more nearly symmetric around zero than simple returns, which are bounded below at negative one but unbounded above. Differencing costs one observation, leaving 6,679 returns per series.

2.4. Sample Period and Structural Context

The extension of the sample beyond the original 2018 endpoint introduces episodes that warrant explicit acknowledgement, since they bear directly on the interpretation of results.

In March and April 2020 the collapse in global oil demand produced the first negative settlement in the history of the West Texas Intermediate futures contract [6] and energy equities fell by more than half within weeks before recovering strongly through 2022. Any parameter estimated on data ending in 2013, as the original study’s trading thresholds are, is unlikely to describe this regime well.

Separately, the identity of the underlying firms changed. Exxon Mobil completed its acquisition of Pioneer Natural Resources in 2024 [7] and ConocoPhillips completed its acquisition of Marathon Oil in the same year [8], in both cases substantially enlarging their Permian Basin positions.

Finally, the weight of the energy sector within the S&P 500 fell to roughly 2% by 2020 before partially recovering, so SPY represents a different mix of economic exposures at the end of the sample than at the beginning. This affects the interpretation of SPY as a control variable but not the internal validity of the pairwise tests among the energy names.

3. EXPLORATORY DATA ANALYSIS

3.1. Descriptive statistics

Table 3.1 reports the moments of the daily log return series. Kurtosis is reported in excess form, so that a Gaussian distribution takes the value zero rather than three.

Table 3.1: Descriptive statistics of daily log returns, 2000 to 2026

	COP	CVX	XOM	XLE	SPY
count	6679.000000	6679.000000	6679.000000	6679.000000	6679.000000
mean	0.000411	0.000373	0.000328	0.000326	0.000312
sd	0.020970	0.017289	0.016555	0.018202	0.012156
min	-0.285552	-0.250062	-0.150271	-0.224910	-0.115887
q25	-0.009715	-0.007875	-0.007784	-0.008277	-0.004607
median	0.000627	0.000838	0.000475	0.000808	0.000693
q75	0.011392	0.009072	0.008836	0.009751	0.005971
max	0.224854	0.204904	0.158631	0.152503	0.135577
skew	-0.492665	-0.472797	-0.089883	-0.651333	-0.208995
kurt	12.658779	19.390471	8.351989	12.210782	11.545178

3.1.1. Central tendency

Mean daily returns lie between 0.031 and 0.041 percent and are economically indistinguishable from one another at daily frequency. Annualising the mean by a factor of 252 gives a clearer picture and annualising the standard deviation by $\sqrt{252}$ permits a crude risk adjustment.

	Annualised mean	Annualised volatility	Ratio
COP	10.4%	33.3%	0.31
CVX	9.4%	27.4%	0.34
XOM	8.3%	26.3%	0.32
XLE	8.2%	28.9%	0.28
SPY	7.9%	19.3%	0.41

Table 3.2: Annualised moments of daily log returns, 1999 to 2026

Two observations follow. The energy names delivered modestly higher returns than the

broad market over the full sample, but did so with substantially greater volatility and on a simple return to volatility basis the broad market dominates all four energy exposures. Separately, the ordering of the ratios among the energy names is tight, spanning 0.28 to 0.34, which is consistent with the securities being driven by a common factor rather than by idiosyncratic performance. This is precisely the condition under which a pairs strategy is plausible, since it implies the securities share a long run driver whose transient dislocations might be tradeable.

The median exceeds the mean for CVX, XOM, XLE and SPY. A median above the mean is the distributional signature of an asset that rises gradually and falls abruptly: the typical day is better than the average day, because the average is dragged down by a small number of severe losses.

3.1.2. Dispersion and volatility ordering

The ordering of standard deviations is COP at 2.10 percent, XLE at 1.82 percent, CVX at 1.73 percent, XOM at 1.66 percent and SPY at 1.22 percent daily. The interpretation is straightforward and economically sensible. SPY is a diversified portfolio of 500 companies across all sectors, so idiosyncratic risk is largely diversified away and only market risk remains. XLE is diversified across energy companies only, so it retains the full sector factor while diversifying firm specific risk, and accordingly sits between the broad market and the individual stocks. Among the single names, COP is the most volatile, which is consistent with it having historically carried the largest relative exposure to upstream production and therefore to the crude price itself.

3.1.3. Skewness

All five series exhibit negative skewness, ordered XLE at -0.65 , COP at -0.49 , CVX at -0.47 , SPY at -0.21 and XOM at -0.09 .

The meaning of negative skewness is frequently misinterpreted as negative days outnum-

bering positive days. As noted above, the median return is positive for four of the five series, so the majority of days are in fact up days. Instead, negative skewness states that the left tail is longer than the right: conditional on a large market move occurring, the large negative moves (downside shocks) are larger in absolute magnitude than the large positive moves (upside rallies). Economically, these assets appreciate through the accumulation of many small gains and depreciate through a small number of large and severe losses. This is the standard structural asymmetry in equity returns and is usually attributed to the interaction of financial leverage, liquidity withdrawal and correlated selling during market stress.

Notably, XOM is close to symmetric at -0.09 , a property that is consequentially observed in the next sub-section below.

3.1.4. Kurtosis and the comparison with the original study

Excess kurtosis is the moment of substantive interest, since it measures the weight in the tails relative to a Gaussian benchmark. Values range from 8.35 for XOM to 19.39 for CVX. For context, a Gaussian distribution has excess kurtosis of exactly zero and values above ten indicate that extreme observations occur at a frequency that a normal model does not remotely accommodate.

Comparing against the original study’s estimates over the shorter 2000 to 2018 window is instructive.

	[1], 2000 to 2018	This study, 2000 to 2026
COP	5.47	12.66
CVX	10.64	19.39
XOM	10.48	8.35
XLE	9.07	12.21
SPY	10.45	11.55

Table 3.3: Excess kurtosis, original sample against extended sample

The direction of these changes is not what mechanical reasoning would predict. Kurtosis is a normalised fourth moment and extending a sample by adding several years of ordinary

trading days ordinarily dilutes it, because the denominator grows while the extreme observations in the numerator stay fixed. However, four of the five series rose and CVX very nearly doubled. That outcome is only possible if the appended period contains observations extreme enough to overwhelm the dilution, which is a quantitative reflection of compounding structural shocks including the historic 2020 COVID crude crash, the 2022 Russia-Ukraine Crisis and ongoing 2026 Middle Eastern supply disruptions.

Exxon Mobil is the exception and the exception is informative. Its excess kurtosis fell from 10.48 to 8.35, the pattern one expects from simple dilution. The explanation is visible in the extreme values. XOM’s minimum daily return of -0.150271 and maximum of 0.158631 are identical to five decimal places to those reported in the original study over the shorter window. Both of XOM’s most extreme days in the full 26 year sample therefore occurred before January 2019 and 2020 did not set a new record for it. COP, CVX, XLE and SPY all set new extremes in the appended period. Whatever the cause, the largest of the majors absorbed the post 2018 dislocations without registering a daily move outside its existing historical range.

3.1.5. The extreme observations in standard deviation units

Dividing each assets series’ minimum daily return by its standard deviation converts the worst observed day into a standardised z -score. This score enables an empirical test of Gaussian model’s validity in describing tail risk.

Table 3.4: Worst observed daily return expressed in standard deviation units (z -scores).

Ticker	Minimum Daily Log Return	Standard Deviations Below Mean (z)
COP	-0.2856	-13.6
CVX	-0.2501	-14.5
XLE	-0.2249	-12.4
SPY	-0.1159	-9.6
XOM	-0.1503	-9.1

These extreme magnitudes are far more than statistical anomalies: the probabilities as-

signed to them by a Gaussian distribution are so infinitesimally small that they lack any physical real world interpretation.

The Mechanics of Gaussian Tail Decay. Standardising a return R_t under the assumption that returns are normally distributed ($R_t \sim N(\mu, \sigma^2)$) gives the standard score:

$$z = \frac{R_t - \mu}{\sigma}, \quad (3.1)$$

The probability of observing a negative shock(loss) at least as severe as k standard deviations corresponds to the lower tail of the standard normal cumulative distribution function (CDF):

$$P(Z \leq -k) = \Phi(-k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-k} e^{-\frac{x^2}{2}} dx. \quad (3.2)$$

For extreme tail events where $k \geq 4$, direct numerical integration causes floating-point underflow in standard computational environments. We therefore apply the asymptotic expansion known as Mills' Ratio:

$$P(Z \leq -k) \approx \underbrace{\frac{1}{k\sqrt{2\pi}}}_{\text{Polynomial scaling}} \times \underbrace{e^{\left(-\frac{k^2}{2}\right)}}_{\text{Super-exponential decay}}, \quad \text{as } k \rightarrow \infty \quad (3.3)$$

Equation (3.3) reveals the core flaw of Gaussian models in finance. Since k is squared inside the negative exponent, probability density decays far faster than a standard exponential curve. Doubling the scale of a market shock ($k \rightarrow 2k$) does not merely quarter its likelihood; it crushes the probability by a factor governed by $e^{-3k^2/2}$.

Worked evaluation: Chevron (*CVX*) at $k = 14.5$. To illustrate the severity of this decay, consider Chevron's minimum daily return, which sits 14.5 standard deviations below its sample mean. Evaluating Equation (3.3) step-by-step:

$$1. \text{ Exponent Evaluation: } \frac{k^2}{2} = \frac{14.5^2}{2} = \frac{210.25}{2} = 105.125 \quad (3.4)$$

$$2. \text{ Exponential Decay Factor: } e^{-105.125} \approx 2.2 \times 10^{-46} \quad (3.5)$$

$$3. \text{ Asymptotic Multiplier: } \frac{1}{k\sqrt{2\pi}} = \frac{1}{14.5 \times 2.5066} \approx 0.0275 \quad (3.6)$$

$$4. \text{ Daily Tail Probability: } P(Z \leq -14.5) \approx 0.0275 \times (2.2 \times 10^{-46}) \approx \mathbf{6 \times 10^{-48}} \quad (3.7)$$

Expected Time. Assuming daily asset returns are independent and identically distributed (i.i.d.), the number of trading days T until a shock of magnitude k occurs follows a discrete Geometric distribution with expected value:

$$E[T] = \frac{1}{p(Z \leq -k)} \quad (3.8)$$

For Chevron's worst day ($k = 14.5$):

$$E[T] \approx \frac{1}{6 \times 10^{-48}} \approx 1.7 \times 10^{47} \text{ trading days} \quad (3.9)$$

Converting trading days into calendar years assuming 252 trading days per year:

$$\text{Expected Years} = \frac{1.7 \times 10^{47}}{252} \approx 6.6 \times 10^{44} \text{ years} \quad (3.10)$$

For context, the age of the observable universe is roughly 1.4×10^{10} years. The Gaussian distribution predicts that a single event like Chevron's worst crash should require 10^{34} universe lifetimes to observe once. Yet, this event appears in a dataset of 6,679 observations (≈ 26 years).

Tail Probabilities Across The Sample. Applying Equation (3.3) across all assets demonstrates that this model breakdown is uniform across market sectors.

	k	$P(Z \leq -k)$ under $N(0, 1)$	Implied waiting time (years)
XOM	9.1	4×10^{-20}	1×10^{17}
SPY	9.6	4×10^{-22}	1×10^{19}
XLE	12.4	1×10^{-35}	3×10^{32}
COP	13.6	8×10^{-43}	5×10^{39}
CVX	14.5	6×10^{-48}	7×10^{44}

Table 3.5: Gaussian tail probabilities and implied waiting times for worst observed returns.

Even ExxonMobil (XOM), the mildest outlier in the sample at -9.1σ , carries a theoretical Gaussian return period that exceeds the age of the universe by seven orders of magnitude.

Violations of the Independence Assumption & Volatility Clustering. The expected waiting time framework assumes independent returns across time. In reality, market volatility exhibits strong autocorrelation, a phenomenon known as volatility clustering, which motivates time-varying variance models such as GARCH. Crucially, volatility clustering invalidates the single-variance Gaussian assumption in a direction that worsens tail risk:

1. When returns transition between low-volatility (σ_{low}) and high-volatility (σ_{high}) regimes, the overall return distribution becomes a mixture of Gaussians.
2. Any mixture of normal distributions with differing variances possesses heavier tails and excess kurtosis than any individual normal component.

Thus, the Gaussian model fails on two independent structural fronts:

- **Functional Form Failure:** Its theoretical tail decay ($e^{-\frac{k^2}{2}}$) drops off too quickly.
- **Homoscedasticity Failure:** Its assumption of a single, static variance (σ) ignores real-world volatility regimes.

Alternative Tail Specification: Heavy-Tailed Power Laws. Financial literature shows that daily stock returns are better described by Power Law distributions, where extreme tail probabilities decay polynomially rather than exponentially:

$$P(|X| > x) \sim C \cdot x^{-\alpha} \quad (3.11)$$

where α represents the tail index (typically estimated between $3 \leq \alpha \leq 4$ for daily US equity returns).

Under a Student's t -distribution with $\nu = 3$ degrees of freedom (which exhibits $\alpha = 3$ asymptotic power-law tail decay), the probability of observing a 14.5σ event is roughly $\sim 10^{-4}$. This corresponds to an expected waiting time of a few decades which is a figure fully consistent with observing it once within a 26-year historical sample.

Comparing the two frameworks:

- Gaussian Model: $P(Z \leq -14.5) \approx 10^{-48} \implies E[T] \approx 10^{44}$ years (Physically impossible)
- Student's t Model ($\nu = 3$): $P(Z \leq -14.5) \approx 10^{-4} \implies E[T] \approx \text{Decades}$ (Empirically possible)

Note on Degrees of Freedom (ν). In parametric tail modeling, the parameter ν does not represent sample size degrees of freedom ($n - 1$). Rather, ν serves as a shape parameter determining tail thickness, where the tail index $\alpha = \nu$. As $\nu \rightarrow \infty$, the Student's t -distribution converges to the standard Gaussian model. Setting $\nu \in [3, 5]$ reflects empirical tail estimates for financial asset returns, capturing finite variance while modeling severe kurtosis.

Implications for Practical Risk Management. Traditional risk metrics like Value-at-Risk (VaR) rely heavily on normality assumptions. For instance, a 99% Gaussian VaR threshold corresponds to 2.33σ . While a Gaussian model predicts that losses will exceed this threshold on 1% of trading days, it is systematically incapable of quantifying the magnitude of losses beyond that point, as it assigns the extreme tail region a probability mass that rounds to zero.

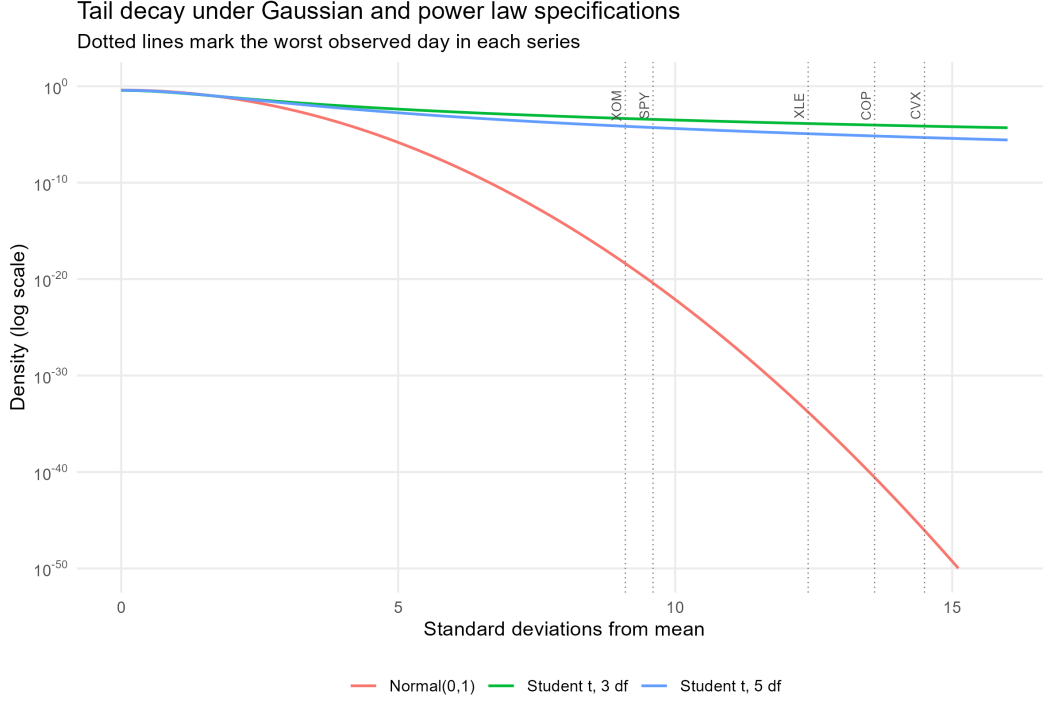


Figure 3.1: Log-density comparison of tail decay under Gaussian vs. heavy-tailed Student's t specifications.

As shown in Figure 3.1, while Gaussian models perform adequately near the distribution center, they collapse in the tail region where survival is determined.

3.2. Formal tests of normality

To complement the empirical tail analysis in Section 3.1.5, Table 3.6 presents hypothesis tests evaluating the Gaussian null hypothesis ($H_0 : R_t \sim \mathcal{N}(\mu, \sigma^2)$). I use the joint Jarque-Bera (JB) omnibus test alongside two decomposed tests: D'Agostino's test for asymmetry (skewness) and the Anscombe-Glynn test for tail thickness (kurtosis).

Every asset series rejects normality at all standard significance levels ($\alpha = 0.01$). The observed JB statistics ranging from 19,421 for XOM to 104,884 for CVX, overshadowing the χ^2_2 critical value of 5.99 at the 5% level by three to four orders of magnitude.

Two nuances govern these results:

1. **Statistical Power vs. Magnitude:** Given $N = 6,679$ observations, hypothesis

Table 3.6: Normality tests on daily log returns

	jb_stat	jb_pval	ad_skew_pval	ansc_kurt_pval	normal
COP	44864.947139	0.000000	0.000000	0.000000	No
CVX	104883.815749	0.000000	0.000000	0.000000	No
XOM	19421.426847	0.000000	0.002734	0.000000	No
XLE	41966.423096	0.000000	0.000000	0.000000	No
SPY	37142.433491	0.000000	0.000000	0.000000	No

tests possess high power to detect trivial non-normalities. Consequently, the rejection itself is expected; the critical finding lies in the *extraordinary scale* of the test statistics relative to their critical threshold.

2. **Moment Decomposition:** Decomposing the JB statistic reveals that while COP, CVX, XLE and SPY reject normality across both moments, ExxonMobil (*XOM*, $p = 0.0027$) is nearly symmetric. XOM’s rejection is driven almost exclusively by extreme kurtosis (ν -driven tail heaviness) rather than skewness.

3.3. Standardised returns and Graphical Analysis

To facilitate direct visual comparison across asset classes, each daily log return series is standardized to zero mean and unit variance ($R_t^* \sim (0, 1)$). A pseudorandom sample drawn from a standard normal distribution ($\mathcal{N}(0, 1)$) of equal sample size ($N = 6,679$) is appended as a baseline control. Since standardisation rescales location and dispersion without modifying higher-order moments, the standardised series preserve exact skewness and kurtosis estimates.

3.3.1. Density comparison

Figure 3.2 overlays the estimated densities of the five standardised return series together with the simulated normal control.

The simulated Gaussian normal peaks at $\frac{1}{\sqrt{2\pi}} \approx 0.399$, confirming expected scaling. In contrast, all empirical series exhibit a pronounced leptokurtic profile:

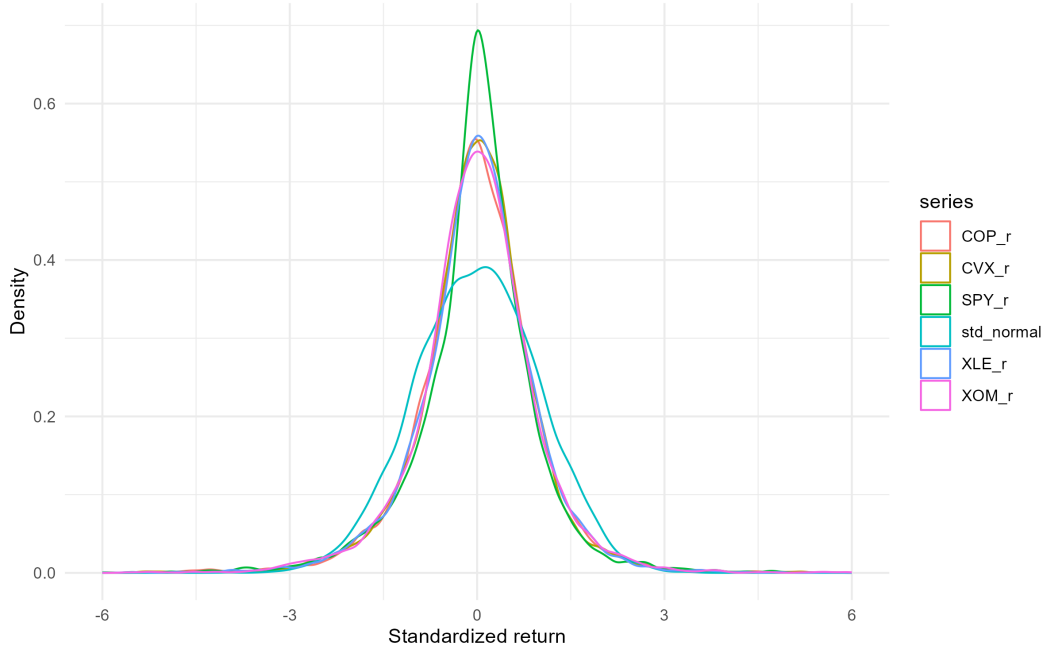


Figure 3.2: Kernel density estimates of standardised daily log returns with a simulated standard normal $\mathcal{N}(0, 1)$ control

- **Elevated Central Peak:** Empirical densities peak markedly higher (0.52 to 0.66), with SPY displaying the highest concentration of returns near zero.
- **Depressed Shoulders:** In the region between 1σ and 2σ , empirical densities fall below the Gaussian control curve.
- **Heavy Tails:** Beyond $|z| > 3$, empirical densities remain elevated where the Gaussian probability density approaches zero.

This structural divergence demonstrates how financial asset returns redistribute variance: relative to a Gaussian baseline, market assets exhibit an excess of near-zero return days, a deficit of moderate fluctuations, and an excess of severe tail shocks.

Because every series possesses unit variance by construction, Figure 3.2 illustrates why variance (σ^2) is an incomplete risk metric: two assets with identical variance can imply fundamentally different probabilities of catastrophic tail events.

3.3.2. Quantile comparison

Figure 3.3 presents normal Quantile-Quantile (Q-Q) plots for each standardised series (R_t^*).

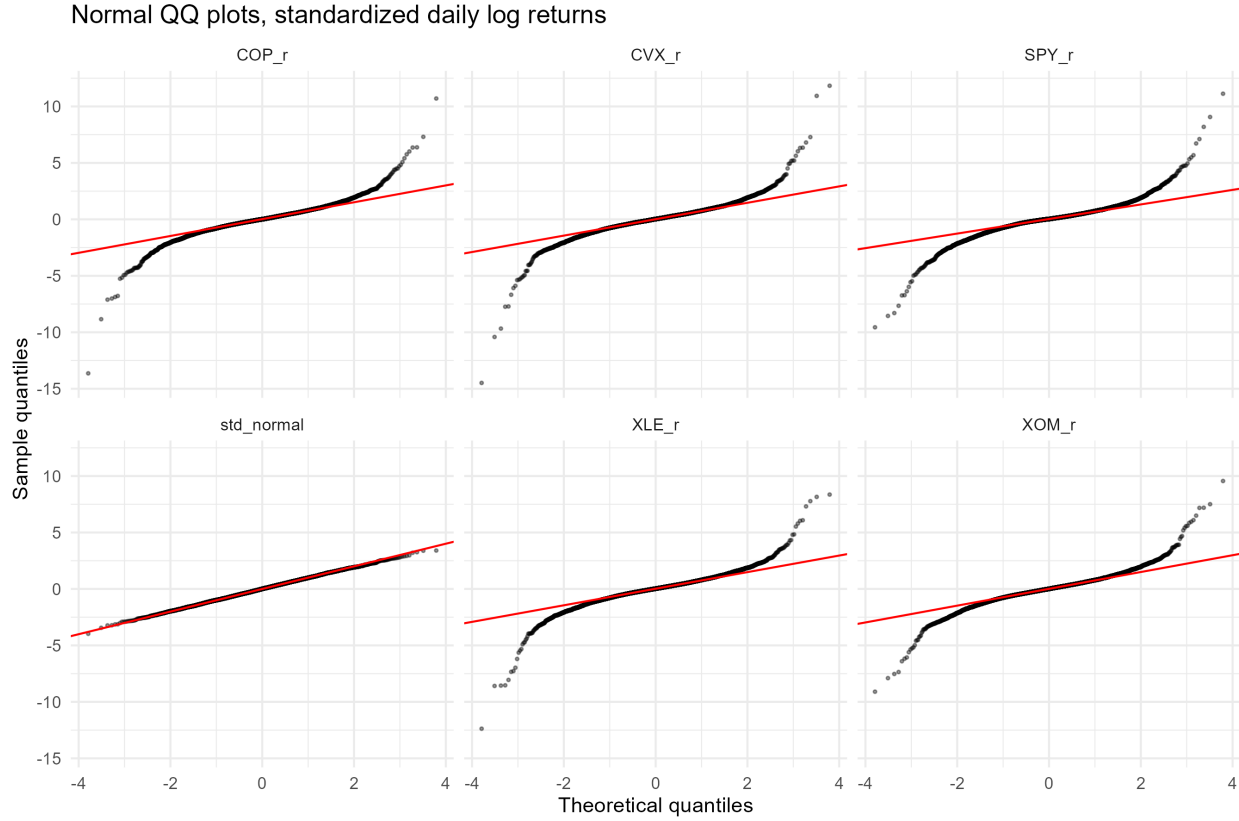


Figure 3.3: Normal Q-Q plots of standardized daily log returns versus theoretical Gaussian quantiles.

A Q-Q plot ranks the observed data and evaluates, for each empirical rank, what value a standard normal distribution would have assigned to it. Under perfect Gaussianity, every observation lies along the 45° reference line ($y = x$). Structural departures are read directionally:

- **Left Tail** ($x < -1.5$): Points falling far below the reference line indicate observed losses significantly more severe than the normal model anticipates.
- **Right Tail** ($x > +1.5$): Points rising far above the reference line indicate observed gains larger than predicted.

Together, these departures form a classic sigmoid pattern, which serves as the visual signature of heavy tails and excess kurtosis. Unsymmetrical deviations across the two tails reflect underlying skewness.

All five empirical series display this sigmoid curvature clearly. The lower tail deviations are noticeably more extreme than the upper tail deviations reaching values between -10σ and -15σ in the left tail compared to $+8\sigma$ to $+12\sigma$ in the right which is consistent with the negative skewness reported in Table 6. Conversely, the simulated normal control adheres strictly to the reference line throughout, confirming that these patterns reflect fundamental market dynamics rather than abstract estimations.

The most revealing feature of these plots is the behavior in the centre. Between approximately -1.5 and $+1.5$ theoretical standard deviations, every empirical series tracks the reference line closely. This demonstrates that while the normal distribution is an adequate description of ordinary trading days, it is severely inadequate for extraordinary ones. Because extreme tail events determine whether risk frameworks, leveraged positions or market-neutral strategies survive, calibrating models on the well-behaved center systematically understates the exact tail risk that matters most.

3.4. Correlation structure

Table 3.7 reports the contemporaneous correlation matrix of daily log returns across the five asset series.

Table 3.7: Correlation matrix of daily log returns					
	COP	CVX	XOM	XLE	SPY
COP	1.0000	0.8110	0.7814	0.8807	0.5520
CVX	0.8110	1.0000	0.8374	0.8931	0.6113
XOM	0.7814	0.8374	1.0000	0.8802	0.5935
XLE	0.8807	0.8931	0.8802	1.0000	0.6592
SPY	0.5520	0.6113	0.5935	0.6592	1.0000

Observations & Interpretation The correlation estimates reveal two distinct patterns in the return data:

1. **Sector Fund Co-Movement:** The energy sector (*XLE*) exhibits the highest overall correlation with each individual integrated major (0.8807 with *COP*, 0.8931 with *CVX*, and 0.8802 with *XOM*). This strong linear relationship is intuitive, as these mega-cap equities constitute primary underlying constituents of the fund.
2. **Inter-Equity Dynamics:** Among individual stocks, the strongest co-movement occurs between Chevron (*CVX*) and ExxonMobil (*XOM*) at 0.8374, reflecting their similar business models and exposure to global integrated oil operations. ConocoPhillips (*COP*) displays strong, albeit slightly lower, correlations with *CVX* (0.8110) and *XOM* (0.7814).
3. **Broad Market Decoupling:** The broad S&P 500 ETF (*SPY*) exhibits noticeably lower correlations with all energy series (0.5520 to 0.6592), highlighting sector-specific factor shocks distinct from overall equity market movements.

4. VECTOR AUTOREGRESSION

4.1. The role of this section

It is worth stating precisely what the vector autoregression is and is not doing here, because the original study is somewhat ambiguous on the point and the ambiguity invites a misreading.

The VAR is not the model that generates trading signals but acts as a screening device. With five securities, there are ten unordered pairs, given by the binomial coefficient:

$$\binom{5}{2} = \frac{5!}{2!(5-2)!} = 10 \quad (4.1)$$

and twenty ordered directional relationships, given by the permutation:

$$P(5, 2) = \frac{5!}{(5-2)!} = 20 \quad (4.2)$$

The VAR framework provides an economically motivated basis for narrowing the candidate set before formal testing begins, by identifying which securities carry short horizon predictive information about one another.

As the VAR is estimated on reuturns to satisfy stationarity requirements, it describes lead-lag dynamics in first differences. Evidence that two return series exhibit mutual predictability indicates local information transmission or a shared short run driver, however, this does not dictate long run equilibrium properties. Therefore, the results of this section are presented strictly as a heuristic filter to reduce the dimensional candidate space. A related but distinct question is being answered here from the one asked in

4.2. Confirming the order of integration

Two tests are applied. The augmented Dickey and Fuller (ADF) test takes the presence of a unit root as its null hypothesis, so that rejection is evidence of stationarity. The Kwiatkowski,

Phillips, Schmidt and Shin (KPSS) test reverses this, taking stationarity as the null, so that rejection is evidence of a unit root. Using both is deliberate. A single test that fails to reject its null is weak evidence, since failure to reject may reflect low power rather than the truth of the null. Two tests with opposing nulls that agree constitute a substantially stronger result as the original study relies on the ADF test alone, mine is improved upon here.

Lag length for the ADF regressions is selected by the Akaike information criterion subject to the upper bound proposed by Schwert,

$$p_{\max} = \left\lfloor 12 \left(\frac{n}{100} \right)^{1/4} \right\rfloor, \quad (4.3)$$

which for $n = 6,680$ gives $p_{\max} = 34$.

Table 4.1: Augmented Dickey and Fuller tests on price levels. Lag length selected by AIC subject to the Schwert bound of 34.

series	adf_stat	cv_1pct	cv_5pct	cv_10pct	lags
COP	-0.0755	-3.43	-2.86	-2.57	30
CVX	0.8095	-3.43	-2.86	-2.57	31
XOM	1.5627	-3.43	-2.86	-2.57	34
XLE	0.6419	-3.43	-2.86	-2.57	23
SPY	3.6481	-3.43	-2.86	-2.57	29

Table 4.2: Augmented Dickey and Fuller tests on daily log returns.

series	adf_stat	cv_1pct	cv_5pct	cv_10pct	lags
COP	-18.8531	-3.43	-2.86	-2.57	19
CVX	-19.5794	-3.43	-2.86	-2.57	21
XOM	-19.8559	-3.43	-2.86	-2.57	17
XLE	-20.1580	-3.43	-2.86	-2.57	17
SPY	-20.3937	-3.43	-2.86	-2.57	15

The results are clear. In levels, the ADF statistics range from -0.08 for ConocoPhillips to $+3.65$ for the S&P 500 fund, all far above the one percent critical value of -3.43 , so the unit root null is not rejected for any series. The corresponding Kwiatkowski, Phillips, Schmidt and Shin statistics range from 38.0 to 47.4 against a one percent critical value of

Table 4.3: Kwiatkowski, Phillips, Schmidt and Shin tests on price levels. The null hypothesis is stationarity.

series	kpss_stat	cv_10pct	cv_5pct	cv_2.5pct	cv_1pct
COP	40.636	0.347	0.463	0.574	0.739
CVX	47.447	0.347	0.463	0.574	0.739
XOM	38.017	0.347	0.463	0.574	0.739
XLE	38.987	0.347	0.463	0.574	0.739
SPY	43.634	0.347	0.463	0.574	0.739

0.739, rejecting stationarity by a factor of approximately 50. In returns, the ADF statistics range from -18.9 to -20.4 , rejecting the unit root decisively in every case. The series are therefore $I(1)$ in levels and $I(0)$ in first differences, which is the configuration required for the analysis that follows.

Two features of the output merit comment.

The ADF statistic for the S&P 500 fund is positive at $+3.65$, which contrasts with ConocoPhillips at -0.08 and may appear anomalous. It is not. A positive statistic indicates an estimated autoregressive coefficient marginally above unity, corresponding to a random walk with positive drift. The fund appreciated from approximately \$92 to \$739 over the sample with a persistent upward drift and this confirms strong nonstationarity driven by a long term economic growth trend.

The elevated lag counts (23 to 34 for levels and 15 to 21 for returns) stem directly from the known large sample mechanics of the AIC [?]. The criterion lacks consistency in lag selection and routinely over estimates lag length in large N environments by preserving terms with trivial marginal fit. Section 4.7 explores this feature further.

Additionally, standard unit root testing can mistake a structural break such as the March 2020 disruption for nonstationarity due to reduced test power. Nevertheless, given the overwhelming test magnitudes and the complete consensus between the ADF and KPSS framework results, the $I(1)$ classification is justified. Formal structural break procedures (e.g., Zivot and Andrews) are unnecessary unless test statistics yield marginal decisions.

4.3. VAR Theoretical Framework

4.3.1. Mathematical Formulation of the $VAR(p)$ Model

To capture the dynamic linear interdependence among the assets, one needs to fomulate a VAR of order p , denoted as $VAR(p)$. Let y_t represent a $K \times 1$ dimensional vector of daily log returns observed at time t , where $K = 5$ corresponds to the the 5 assets in this paper:

$$y_t = \begin{bmatrix} R_{COP,t} \\ R_{CVX,t} \\ R_{XOM,t} \\ R_{XLE,t} \\ R_{SPY,t} \end{bmatrix} \quad (4.4)$$

Rhe general system equation for observation t is expressed as:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + e_t \quad (4.5)$$

where c is a $K \times 1$ vector of constant intercept terms:

$$c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_K \end{bmatrix} \quad (4.6)$$

Each A_l (for $l = 1, \dots, p$) is a $K \times K$ autoregressive coefficient matrix regulating the transmission of returns from lag l to time t :

$$A_l = \begin{bmatrix} a_{1,1}^{(l)} & a_{1,2}^{(l)} & \dots & a_{1,k}^{(l)} \\ a_{2,1}^{(l)} & a_{2,2}^{(l)} & \dots & a_{2,k}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(l)} & a_{K,2}^{(l)} & \dots & a_{K,K}^{(l)} \end{bmatrix} \quad (4.7)$$

The diagonal elements $a_{i,i}^{(l)}$ capture direct own-lag return persistence, while the off diagonal elements $a_{i,j}^{(l)}$ ($i \neq j$) capture cross asset lead lag predictability and spillover dynamics across time horizons.

The vector $e_t = [e_{1,t}, e_{2,t}, \dots, e'_{K,t}]$ denotes unobservable $K \times 1$ white noise innovations occurring at time t . The innovation (shock) process satisfies:

$$\mathbb{E}[e_t] = 0 \quad (4.8)$$

$$\mathbb{E}[e_t e'_s] = \begin{cases} \Omega, & \text{if } t = s \\ 0, & \neq s \end{cases} \quad (4.9)$$

where Ω represents the $K \times K$ contemporaneous innovation covariance matrix:

$$\Omega = \mathbb{E}[e_t e'_t] = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \dots & \sigma_{1,K} \\ \sigma_{2,1} & \sigma_2^2 & \dots & \sigma_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{K,1} & \sigma_{K,2} & \dots & \sigma_K^2 \end{bmatrix} \quad (4.10)$$

where the diagonal elements $\sigma_i^2 = \text{Var}(e_{i,t})$ denote the unforecastable variance intrinsic to asset i and the off diagonal elements $\sigma_{i,j} = \text{Cov}(e_{i,t}, e_{j,t})$ for $i \neq j$ measure contemporaneous shock co-movements between asset i and asset j occurring within the same trading period t .

Because Ω is a 5×5 symmetric matrix and positive definite, the implied contemporaneous residual correlation between asset innovations is defined as:

$$\rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j} \quad (4.11)$$

While the coefficient matrices $A_1, \dots, A_p$ capture dynamic intertemporal lead-lag linkages, Ω isolates contemporaneous joint shocks driven by unmodeled sector-wide or macroeconomic news arrivals.

Expanding the system into full matrix notation yields the complete structural representation for observation t :

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ \vdots \\ y_{K,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_K \end{bmatrix} = \begin{bmatrix} a_{1,1}^{(1)} & a_{1,2}^{(1)} & \dots & a_{1,K}^{(1)} \\ a_{2,1}^{(1)} & a_{2,2}^{(1)} & \dots & a_{2,K}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(1)} & a_{K,2}^{(1)} & \dots & a_{K,K}^{(1)} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ \vdots \\ y_{K,t-1} \end{bmatrix} + \dots + \begin{bmatrix} a_{1,1}^{(p)} & a_{1,2}^{(p)} & \dots & a_{1,K}^{(p)} \\ a_{2,1}^{(p)} & a_{2,2}^{(p)} & \dots & a_{2,K}^{(p)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(p)} & a_{K,2}^{(p)} & \dots & a_{K,K}^{(p)} \end{bmatrix} \begin{bmatrix} y_{1,t-p} \\ y_{2,t-p} \\ \vdots \\ y_{K,t-p} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \\ \vdots \\ e_{K,t} \end{bmatrix} \quad (4.12)$$

4.4. Stacked Matrix System Representation

To transition from equation by equation representations to a computationally efficient estimation framework over a sample period of T trading days, we collect daily observation vectors into consolidated block matrices.

Because a $\text{VAR}(p)$ model requires p past observations as historical inputs, to generate its first prediction, the first p time periods cannot be estimated. Consequently, the actual estimation begin at day $t = p + 1$ and spans an effective sample size of $T^* = T - p$ time steps (from $t = p + 1$ through T). The entire multi equation time series system is compactly expressed as:

$$Y = BZ + U \quad (4.13)$$

where Y , B , Z and U are partitioned matrices structured to align temporal and cross

sectional dimensions:

1. **Dependent Variable Matrix (Y):** The matrix $Y \in \mathbb{R}^{K \times (T-p)}$ collects observed daily return vectors side by side across time steps $t = p + 1$ through T :

$$Y = \begin{bmatrix} | & | & & | \\ y_{p+1} & y_{p+2} & \cdots & y_T \\ | & | & & | \end{bmatrix} = \begin{bmatrix} y_{1,p+1} & y_{1,p+2} & \cdots & y_{1,T} \\ y_{2,p+1} & y_{2,p+2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{K,p+1} & y_{K,p+2} & \cdots & y_{K,T} \end{bmatrix} \quad (4.14)$$

2. **Parameter Matrix (B):** The parameter matrix $B \in \mathbb{R}^{k \times (1+Kp)}$ horizontally concatenates the constant intercept vector c and the sequence of p autoregressive coefficient matrices $[c \ A_1 \ \dots \ A_p]$:

$$B = \begin{bmatrix} c_1 & a_{1,1}^{(1)} & \cdots & a_{1,K}^{(1)} & \cdots & a_{1,1}^{(p)} & \cdots & a_{1,K}^{(p)} \\ c_2 & a_{2,1}^{(1)} & \cdots & a_{2,K}^{(1)} & \cdots & a_{2,1}^{(p)} & \cdots & a_{2,K}^{(p)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_K & a_{K,1}^{(1)} & \cdots & a_{K,K}^{(1)} & \cdots & a_{K,1}^{(p)} & \cdots & a_{K,K}^{(p)} \end{bmatrix} \quad (4.15)$$

3. **Regressor Matrix (Z):** The regressor matrix $Z \in \mathbb{R}^{(1+Kp) \times (T-p)}$ aggregates the explanatory variables. Column t of Z contains the constant intercept unit alongside all lagged return vectors required to predict column t of Y :

$$Z = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ y_p & y_{p+1} & \cdots & y_{T-1} \\ y_{p-1} & y_p & \cdots & y_{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ y_1 & y_2 & \cdots & y_{T-p+1} \end{bmatrix} \quad (4.16)$$

4. **Residual Matrix (U):** The matrix $U \in \mathbb{R}^{K \times (T-p)}$ collects unobserved innovation vectors e_t for each estimated time step:

$$U = \begin{bmatrix} | & | & & | \\ e_{p+1} & e_{p+2} & \dots & e_T \\ | & | & & | \end{bmatrix} = \begin{bmatrix} e_{1,p+1} & e_{1,p+2} & \dots & e_{1,T} \\ \vdots & \vdots & \ddots & \vdots \\ e_{K,p+1} & e_{K,p+2} & \dots & e_{K,T} \end{bmatrix} \quad (4.17)$$

4.5. Closed Form System Estimation via OLS

Every individual equation in the VAR(p) system features an identical design matrix Z on the right-hand side. Because the explanatory variables are identical across all K assets, the Seemingly Unrelated Regression (SUR) system estimator collapses directly into equation by equation Ordinary Least Squares (OLS) without loss of asymptotic efficiency.

To solve for the parameter matrix B , we post-multiply the stacked system $Y = BZ + U$ by the transpose of the regressor matrix, Z' :

$$YZ' = B(ZZ') + UZ' \quad (4.18)$$

The matrix inner product $(ZZ') \in \mathbb{R}^{(1+Kp) \times (1+Kp)}$ transforms the rectangular regressor array into a symmetric, square sample autocovariance matrix. Under the standard Gauss-Markov exogeneity assumption ($\mathbb{E}[UZ'] = 0$), therefore the expected cross-product of lagged regressors and unforecastable innovations cancels out. Dropping the residual term leaves the system of Normal Equations:

$$YZ' = B(ZZ') \quad (4.19)$$

Multiplying both sides by the matrix inverse $(ZZ')^{-1}$ isolates the closed-form OLS parameter estimator $\hat{B}$:

$$\hat{B} = YZ'(ZZ')^{-1} \quad (4.20)$$

This closed-form result means that estimating a system of K equations containing $K(1 + Kp)$ total parameters requires only a single matrix inversion of dimension $(1 + Kp) \times (1 + Kp)$, bypassing iterative optimization routines.

4.6. Residual Extraction and Innovation Covariance Estimation

Once $\hat{B}$ is estimated, we evaluate fitted values $\hat{Y} = \hat{B}Z$ and extract the empirical sample residual matrix $\hat{U}$:

$$\hat{U} = Y - \hat{B}Z \quad (4.21)$$

Column t of $\hat{U}$ yields the estimated $K \times 1$ innovation vector $\hat{e}_t = [\hat{e}_{1,t}, \dots, \hat{e}_{K,t}]'$.

To estimate the contemporaneous innovation covariance matrix $\Omega = \mathbb{E}[e_t e_t']$, we compute the cross product matrix of empirical residuals, $\hat{U}\hat{U}'$:

$$\hat{U}\hat{U}' = \sum_{t=p+1}^T \hat{e}_t \hat{e}_t' \quad (4.22)$$

To ensure that $\hat{\Omega}$ is an unbiased estimator, we adjust the scale factor for loss of degrees of freedom. The effective sample size is $T^* = T - p$, and each equation in the VAR system consumes $1 + Kp$ degrees of freedom to estimate its parameter vector. The unbiased estimator for the $K \times K$ innovation covariance matrix is defined as:

$$\hat{\Omega} = \frac{1}{(T - p) - (1 + Kp)} \hat{U}\hat{U}' \quad (4.23)$$

4.7. Lag order selection

Determining the autoregressive lag order p requires balancing model fit against overparameterisation. Choosing an excessively large p risks overfitting in sample noise and inflating estimator variance. On the other hand selecting a lag order that is too small introduces omitted variable bias.

The lag order p is formally selected by minimising a Information Criterion $IC(p)$ evaluated across a candidate grid $p \in \{1, 2, \dots, p_{\max}\}$. Each criterion balances a goodness-of-fit metric which is measured by the log determinant of the estimated innovation covariance matrix $\ln |\hat{\Omega}(p)|$ against a penalty term that increases with the total number of system parameters pK^2 :

$$\text{AIC}(p) = \ln |\hat{\Omega}(p)| + \frac{2}{T}pK^2, \quad (4.24)$$

$$\text{HQ}(p) = \ln |\hat{\Omega}(p)| + \frac{2 \ln \ln T}{T}pK^2, \quad (4.25)$$

$$\text{SC}(p) = \ln |\hat{\Omega}(p)| + \frac{\ln T}{T}pK^2. \quad (4.26)$$

Akaike's Final Prediction Error (FPE) criterion, which is asymptotically equivalent to the AIC in moderate and large samples, is evaluated alongside AIC for completeness:

$$\text{FPE}(p) = \left(\frac{T + Kp + 1}{T - Kp - 1} \right)^K |\hat{\Omega}(p)| \quad (4.27)$$

When evaluated on the empirical sample ($T = 6,677$, $K = 5$) across a maximum lag search space of $p_{\max} = 20$, the criteria yield conflicting choices, as reported in Table 4.4.

	AIC	HQ	SC	FPE
Selected order	9	1	1	9

Table 4.4: Lag order selected by each Information Criterion, $p_{\max} = 20$

The disagreement is structural rather than incidental and arises directly from the mag-

nitude of the parameter penalty relative to the sample size. At $T = 6,677$, the Schwarz (SC/BIC) penalty carries a factor of $\ln T \approx 8.81$ per parameter, compared to the fixed Akaike (AIC) factor of 2. SC/BIC is therefore facing a tax ratio exceeding four to one. Because each additional lag introduces $K^2 = 25$ new coefficients across the system, the two criteria price model expansion at fundamentally different rates. The Hannan-Quinn (HQ) and Schwarz (SC) criteria are consistent estimators that converge to the true lag order as $T \rightarrow \infty$, whereas AIC and FPE are known to over select asymptotically in large samples.

4.7.0.1. Mathematical Inspection of AIC To explicitly demonstrate the marginal nature of the AIC preference for $p = 9$, we evaluate the criterion values across $p = 1$ and $p = 9$. Given $T^* = 6,677$ and $K = 5$ ($K^2 = 25$ parameters per lag), the criterion values evaluate as follows:

$$\text{AIC}(1) = \ln |\hat{\Omega}(1)| + \frac{2}{6,677}(25) = -46.1791 \quad (4.28)$$

$$\text{AIC}(9) = \ln |\hat{\Omega}(9)| + \frac{2}{6,677}(225) = -46.1977 \quad (4.29)$$

The absolute reduction in the criterion metric is:

$$\Delta \text{AIC} = 46.1977 - 46.1791 = 0.0186 \quad (4.30)$$

Scaling this reduction against the base metric at $p = 1$ yields the relative gain:

$$\text{Relative Gain} = \frac{0.0186}{46.1791} \approx 0.0004027 \quad \left(\text{or } \approx \frac{4}{10,000} \right) \quad (4.31)$$

Estimating 200 additional parameters yields a 0.04% relative improvement in fit, confirming that $p = 9$ captures sampling noise rather than macroeconomic structure.

4.7.1. Graphical Analysis the Chosen Information Criteria



Figure 4.1: AIC, HQ and SC evaluated across lag orders $p \in \{1, \dots, 20\}$. The open circles highlight the global minimum of each Information Criteria.

As illustrated in Figure 4.1, the smoothed AIC surface displays an essentially flat profile beyond the initial lags. Although the metric reaches a technical global minimum at $p = 9$, the curve reflects a near-zero slope across the entire candidate space $p \in [1, 10]$. This smooth, horizontal plateau confirms that incremental lags beyond $p = 1$ yield negligible information gains, causing AIC to slowly drift toward a shallow minimum driven by high-frequency sample variation rather than strong dynamic feedback. By contrast, the Schwarz criterion increases monotonically for all $p > 1$, reflecting a unambiguous preference for model parsimony.

Neither criterion selects $p = 2$, which was the order specified in the baseline study. Rather than adopting an ad-hoc composite ranking across criteria, this paper maintains $p = 2$ as the primary specification purely to ensure direct comparability with the benchmark literature. However, acknowledging that statistical criteria over this extended sample favor $p = 1$, we explicitly report the VAR(1) specification as a primary robustness check.

The economic rationale for parsimony reinforces this statistical choice: a VAR(9) specification requires estimating 225 autoregressive parameters to model daily asset returns, where the predictable dynamic component accounts for roughly one percent of total variance. A sparse baseline model ($p \in \{1, 2\}$) preserves estimation precision and prevents out of sample forecast degradation.

4.8. Estimation Results

Table 4.5 reports the estimated coefficients of the VAR(2) system fitted to 6,677 observations across five assets: ConocoPhillips (COP), Chevron (CVX), ExxonMobil (XOM), Energy Select Sector SPDR (XLE), and S&P 500 ETF (SPY).

Table 4.5: VAR(2) Coefficient Estimates and Model Fit Statistics					
	COP	CVX	XOM	XLE	SPY
COP.L1	-0.0434.	-0.0334	-0.0447*	-0.0051	-0.0188
CVX.L1	-0.1447***	-0.1249***	-0.0826**	-0.1087***	-0.0514**
XOM.L1	0.0428	0.0173	0.0110	0.0462	0.0107
XLE.L1	0.1153*	0.1150**	0.0856*	0.0257	0.0249
SPY.L1	-0.0703*	-0.0942***	-0.1077***	-0.0259	-0.0572***
COP.L2	-0.0094	0.0197	0.0046	0.0119	0.0123
CVX.L2	-0.0154	-0.0019	0.0063	-0.0197	0.0148
XOM.L2	-0.0669*	-0.0814**	-0.0998***	-0.0785**	-0.0365.
XLE.L2	0.0353	0.0210	0.0251	0.0246	0.0137
SPY.L2	0.0618*	0.0477*	0.0260	0.0404	-0.0354*
const	0.0005.	0.0004*	0.0004*	0.0004.	0.0004*
R^2	0.0078	0.0116	0.0148	0.0067	0.0111
Adj. R^2	0.0063	0.0101	0.0133	0.0052	0.0096
F statistic	5.21	7.82	10.02	4.48	7.49
Resid. s.e.	0.02090	0.01720	0.01644	0.01815	0.01209

Notes: Dependent variables are daily returns. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.10$.

4.8.1. Goodness of Fit and Predictive Capacity

Coefficients of determination (R^2) range from 0.0067 for the energy sector fund to 0.0148 for Exxon Mobil. The system explains under one and a half percent of the variation in daily

returns in every equation.

The F -statistics are nevertheless significant at any conventional level, ranging from 4.48 to 10.02 with p -values below 10^{-5} throughout. There is therefore genuine predictability in the statistical sense. What there is not, however, is predictability of an economically meaningful magnitude.

This divergence highlights a critical distinction between statistical and economic significance:

1. **Statistical Power Inflation:** With $n = 6,677$ daily observations, sample standard errors shrink drastically ($\approx 1/\sqrt{n}$). Consequently, the system possesses sufficient power to reject the null hypothesis for minute coefficient estimates, yielding overwhelming statistical significance ($p < 10^{-5}$) for relationships far too weak to be profitably exploited.
2. **Near-Efficient Market Dynamics:** In highly liquid equity markets, obvious cross-asset lead-lag dependencies are rapidly arbitrated away. The remaining $\approx 1\%$ signal reflects residual microstructure friction rather than unexploited alpha.

This observation bounds what the remainder of the paper can claim; a methodological limitation not addressed in the original study.

4.8.2. Stability

A $\text{VAR}(p)$ system is dynamically stable if all eigenvalues of its companion matrix lie strictly within the complex unit circle, that is, if the roots of the characteristic polynomial $|\lambda I - C| = 0$ satisfy $|\lambda| < 1$.

To formulate this, consider a K variable $\text{VAR}(p)$ process,

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + u_t. \quad (4.32)$$

To evaluate dynamic stability, the p lag system is mapped into an equivalent Kp dimen-

sional VAR(1) companion form,

$$Z_t = CZ_{t-1} + U_t, \quad (4.33)$$

where

$$Z_t \equiv \begin{bmatrix} Y_t \\ Y_{t-1} \\ \vdots \\ Y_{t-p+1} \end{bmatrix}, \quad U_t \equiv \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad (4.34)$$

and C is the $Kp \times Kp$ companion matrix

$$C = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}. \quad (4.35)$$

For a 5-variable VAR(2) model ($K = 5, p = 2$), C is a 10×10 matrix:

$$C = \begin{bmatrix} A_1 & A_2 \\ I_5 & 0_5 \end{bmatrix} \quad (4.36)$$

The computed root moduli for this system are given by:

$$\{|\lambda_i|\}_{i=1}^{10} = \{0.256, 0.256, 0.243, 0.243, 0.167, 0.167, 0.144, 0.144, 0.135, 0.135\}. \quad (4.37)$$

All roots lie comfortably within the unit circle, with the maximum root modulus bounded at $\max_i |\lambda_i| \approx 0.256 < 1$. This satisfies the necessary and sufficient conditions for stationarity and dynamic stability. Consequently, the vector moving average representation, VMA(∞), required for the impulse response functions exists, is uniquely identified and con-

verges.

The appearance of roots in exact duplicate pairs indicates complex conjugate eigenvalues, $\lambda_i = a \pm bi$. Economically, this implies that systemic decay follows an oscillatory path rather than a monotonic dampening, aligning with the negative first order lag coefficients documented in Section 4. Furthermore, the low maximum modulus ($\max|\lambda| \approx 0.256$) shows a very weak shock persistence in the system; a unit disturbance dissipates by over 93% within two trading days ($0.256^2 \approx 0.065$), confirming rapid information digestion and return to equilibrium. (See Appendix ?? for companion matrix formulation)

4.8.3. Coefficient sign pattern and a microstructure caveat

An examination of the coefficient signs reveals a pattern that warrants careful interpretation and that the original study does not address.

The first lag of Chevron (CVX) enters negatively and significantly in all five equations, with coefficients from -0.051 to -0.145 . Similarly, the first lag of the S&P 500 ETF fund (SPY) enters negatively in all five (reaching statistical significance in three). Own first lags are consistently negative wherever statistically significant. Furthermore, the second lag of Exxon Mobil (XOM) enters negatively and significantly in four of five equations.

Negative first order autocorrelation in daily equity returns is a well documented artefact of market microstructure rather than evidence of exploitable mean reversion in the efficient price. The primary driver is bid-ask bounce : consecutive market executions alternate between bid quotes (P_{bid}) and ask quotes (P_{ask}), generating spurious negative serial correlation in observed transaction returns even when efficient price follows a martingale process. Because this study uses adjusted closing prices, which are transaction prices, the effect is present by construction.

To see this formally, consider an asset whose true unobserved equilibrium price P_t^* follows a random walk:

$$P_t^* = P_{t+1}^* + \epsilon_t \tag{4.38}$$

Due to the spread $S = P_{\text{ask}} - P_{\text{bid}}$, observed transaction prices P_t are recorded as:

$$P_t = P_t^* + I_t \left(\frac{S}{2} \right) \quad (4.39)$$

where $I_t \in \{+1, -1\}$ is an independent order flow indicator determining whether a trade executed at the ask (+1) or the bid (-1). Taking the first differences to calculate return yields an observed return series $r_t = \Delta P_t$:

$$r_t = \varepsilon_t + (I_t - I_{t-1}) \left(\frac{S}{2} \right) \quad (4.40)$$

Computing the first-order autocovariance of these observed returns reveals a negative bias induced directly by the spread noise:

$$\text{Cov}(r_t, r_{t-1}) = -\text{Var}(I_t) \left(\frac{S}{2} \right)^2 < 0 \quad (4.41)$$

Because this study relies on daily adjusted closing prices which are derived from execution prices this mechanical transaction noise is present by construction.

Consequently, the predictive relationships identified by the VAR merge two distinct phenomena: genuine economic lead lag dynamics across energy equities and mechanical noise introduced by the market clearing mechanism. Because transaction noise is not explicitly filtered here, the estimated VAR model should be interpreted strictly as an exploratory screening heuristic rather than a source of tradeable execution signals. Re-estimating the system using quote midpoints $\left(P_{\text{mid}} = \frac{P_{\text{ask}} + P_{\text{bid}}}{2} \right)$ rather than transaction prices represents the natural extension to isolate true fundamental price discovery.

4.8.4. Innovation covariance structure

The correlation matrix of residuals is reported in Table 4.6.

Contemporaneous residual correlations among the three integrated majors and the sector fund are high, between 0.78 and 0.89, with the sector fund most strongly correlated with

Table 4.6: Contemporaneous correlation matrix of VAR(2) residuals.

	COP	CVX	XOM	XLE	SPY
COP	1.0000	0.8097	0.7805	0.8804	0.5506
CVX	0.8097	1.0000	0.8363	0.8937	0.6095
XOM	0.7805	0.8363	1.0000	0.8808	0.5899
XLE	0.8804	0.8937	0.8808	1.0000	0.6588
SPY	0.5506	0.6095	0.5899	0.6588	1.0000

each individual name. Correlations involving the broad market fund are markedly lower, between 0.55 and 0.66.

Two implications follow. First, the broad market fund is distinguishable from the four energy exposures in a way that supports its treatment as a control rather than as a candidate pair leg. Second, and more consequentially, the high correlations among the energy series mean that the innovations are far from orthogonal i.e (Correlation = 0). When the contemporaneous correlation between two innovations is 0.89, the attribution of a common movement to one series or the other is close to arbitrary, and any procedure requiring such an attribution must be examined for sensitivity to the assumption it imposes.

A comparison with the original study highlights how sample extension across volatile macro regimes alters error covariance structure. In the original sample, the contemporaneous residual correlations between the sector benchmark (XLE) and the individual majors (COP, CVX and XOM) were recorded at 0.855, 0.878 and 0.854, respectively. In our extended sample, these parameters shift upward to 0.880, 0.894 and 0.881.

This spike in residual co movement aligns directly with the empirical property of asymmetric cross asset correlation during systemic downturns. As established in Section 3.1.5, the extended sample incorporates the unprecedented energy market volatility of 2020, a period characterized by simultaneous crude demand destruction and forced institutional portfolio deleveraging. During such tail events, asset prices cease to trade on company-specific idiosyncratic fundamentals and instead respond synchronously to macro-level liquidity shocks. Because covariance calculations are heavily weighted by extreme realizations, these clustered

outliers pull the overall linear dependency upward, rendering the identification problem even more acute in the post-2020 sample than in the original benchmark.

APPENDIX SECTION

A.1. Full lag order selection criteria

Table A.1 reports the value of each information criterion for every lag order from 1 to 20, underlying the selection discussion in Section 4.7.

Table A.1: Information criteria by lag order, $p = 1$ to 20				
Lag	AIC	HQ	SC	FPE
1	-46.179083	-46.168494	-46.148431	0.0000000000
2	-46.183745	-46.164332	-46.127549	0.0000000000
3	-46.184343	-46.156107	-46.102605	0.0000000000
4	-46.190426	-46.153365	-46.083144	0.0000000000
5	-46.191495	-46.145610	-46.058669	0.0000000000
6	-46.191438	-46.136730	-46.033069	0.0000000000
7	-46.197123	-46.133590	-46.013211	0.0000000000
8	-46.196042	-46.123685	-45.986586	0.0000000000
9	-46.197694	-46.116513	-45.962695	0.0000000000
10	-46.197373	-46.107369	-45.936831	0.0000000000
11	-46.195806	-46.096978	-45.909720	0.0000000000
12	-46.193248	-46.085595	-45.881619	0.0000000000
13	-46.193829	-46.077353	-45.856657	0.0000000000
14	-46.189325	-46.064025	-45.826609	0.0000000000
15	-46.197227	-46.063102	-45.808968	0.0000000000
16	-46.194847	-46.051899	-45.781045	0.0000000000
17	-46.193316	-46.041544	-45.753971	0.0000000000
18	-46.192856	-46.032260	-45.727967	0.0000000000
19	-46.189285	-46.019865	-45.698853	0.0000000000
20	-46.187199	-46.008955	-45.671223	0.0000000000

Bibliography

- [1] C. Salas Najera, “Pair trading and VAR analysis applied to energy stocks,” tech. rep., Certificate in Quantitative Finance, June 2019.
- [2] D. Goldman, “Oil market faces biggest threat since start of Iran war.” CNN Business, July 2026.
- [3] R. Bookstaber, A Demon of Our Own Design: Markets, Hedge Funds, and the Perils of Financial Innovation. Wiley, 2006.
- [4] ConocoPhillips, “Conocophillips becomes one of the world’s largest independent e&p companies, spinoff of phillips 66 complete.” <https://www.conocophillips.com/news-media/story/conocophillips-becomes-one-of-the-worlds-largest-independent-ep-companies-spinoff-of-2012>. 2012.
- [5] Center for Research in Security Prices, “Chapter 5: Crsp calculations,” tech. rep., University of Michigan, 2023.
- [6] U.S. Energy Information Administration, “Crude oil prices briefly traded below \$0 in spring 2020 but have since been mostly flat.” <https://www.eia.gov/todayinenergy/detail.php?id=46336>, 2020.
- [7] Exxon Mobil Corporation, “Exxonmobil completes acquisition of pioneer natural resources.” https://corporate.exxonmobil.com/news/news-releases/2024/0503_exxonmobil-completes-acquisition-of-pioneer-natural-resources, 2024.
- [8] The Energy Year, “Conocophillips completes acquisition of marathon oil.” <https://theenergyyear.com/news/conocophillips-completes-acquisition-of-marathon-oil/>, 2024.